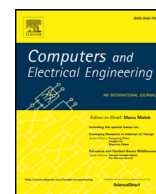




Contents lists available at ScienceDirect

Computers and Electrical Engineering

journal homepage: www.elsevier.com/locate/compelecengCrop yield forecasting using fuzzy logic and regression model[☆]

Bindu Garg, Shubham Aggarwal*, Jatin Sokhal

Department of Computer Science & Engineering, Bharati Vidyapeeth's College of Engineering, New Delhi 110063, India

ARTICLE INFO

Article history:

Received 10 March 2017

Revised 11 November 2017

Accepted 11 November 2017

Available online xxx

Keywords:

Mean Square Error

Fuzzy time series

Average forecast error rate

Prediction

Wheat yield

Time series data

ABSTRACT

Time-Series data has been of great importance to the research field of prediction models. Over the past two decades, multitudinous fuzzy time series blueprints have been put forth for agricultural yield production. However, most of these predictions were based on 7th interval partitioning. A surprising insight was that nobody gave a sound reason to justify the choice of that particular interval. So, this paper focuses on predicting data values on a large spectrum of fuzzy logic computations based on second and third-degree relationships. This paper showcases work on 4 different types of the fuzzy interval, where each interval is tested with 4 degrees of regression equations. Each of these 16 cases is performed for the fuzzy logic relationship (FLR) 2 and 3 separately. Apart from this, the robustness of algorithm is a testament to an incredible solution for the time series model. In addition to this, A Regression analysis model has been enforced to accomplish the efficient defuzzification operation. To elucidate the process of forecasting, the historical data of wheat yield of University of Agriculture and Technology has been used.

© 2017 Elsevier Ltd. All rights reserved.

1. Introduction

The forecasting methodology is very befitting in the cases where incertitude related to the future is concrete. The prediction of outcomes in the future is attained through this process. Pertinent data and graphs are deliberated and queried in order to make optimal choices concerning the future. The employment of time series forecasting has come into picture predominantly for two reasons. First, time series data forms a preponderant part of the data existing in business, economic and financial areas. Next, it is indeed facile to appraise time series as many technologies are procurable for evaluation of time series forecast.

The inspiration for this research work came from the authors' previously published research work [24] and study of previous research work conducted in the area of predictive modeling using fuzzy logic. Various authors in the past have used a particular interval of partitioning in their work but have not justified non-inclusion of other intervals. Most researchers used 7th interval partitioning in their research work but have not justified non-inclusion of other intervals. Use of this particular partition by all made the authors of this paper curious to find out how the results vary by changing the interval of partitioning and even within one interval and how the prediction values change with the change in polynomial degree equation

[☆] Reviews processed and recommended for publication to the Editor-in-Chief by Guest Editor Dr. R. C. Poonia.

* Corresponding author.

E-mail address: shubhamaggrwl@gmail.com (S. Aggarwal).

in regression analysis. A substantial segment of the work on time series has been performed to address and unravel the solutions to problems like a number of outpatient visits, healthcare, prediction in information systems forecasting, economic and sales forecasting, analysis of the budget, stock market prediction and fluctuations and business analysis etc. Thus, there exists a persistent demand for forecasting techniques that offer optimal and veracious results. Thus, the need of the hour is the demand of soft computing based forecasting methods which are efficacious and consummate.

This paper explains the novel contribution to the readers in an easy to navigate sections. Section 1 gives a brief introduction on the topic of fuzzy logic and the inspiration for this paper. Section 2 explains the significance of wheat production is explained in brief. In Section 3, literature survey has been done in order to let users understand the progress in soft computing field in the past. Section 4 presents the proposed method is easy to understand steps along with tables to understand the data distribution. The authors have further subdivided the proposed method into four sections in order to show the results of the application algorithm on polynomial degrees 1 to 4. In Section 5, the results obtained are explained through graphical illustrations in order to give a quick glance of comparison among different intervals. Section 6 shows the robustness of the algorithm to substantiate the algorithm's strength. In Section 7, the paper concludes with discussion on future scope of this paper.

2. Significance of wheat production

India is one of the largest producers of wheat in the world (close to one – fifth of the total wheat yield worldwide). Wheat is the staple diet of the people residing in the eastern and southern parts of the country. Also, it is one of the chief grains of the nation and accounts for the largest portion of the agricultural area under its cultivation. Wheat, being a tropical plant, burgeons comfortable in hot climate and India, a tropical country, endorses this argument. Farming of the wheat crop is considered as a very lucrative, remunerative and sustainable venture throughout the world. The Asian continent on its own produces and consumes more than three-fourths of net wheat production. If sources are to be believed, the recent sharp boom in the Asian wheat production will act as a means to lessen poverty. Since improved production and growth in yield of wheat clearly infers that wheat would be easily available to poor people at a lower price.

Henceforth, the relatively reduced cost of wheat will result in incitation of farmers to invest in higher valued crops and thus, bring prosperity and added income to their families, which will also secure enhanced nutrition for the customers. The evaluation and prediction of wheat production is indeed an Augean field. Past work done on wheat bespeaks that there exists a variation in wheat production, which in addition to being really unpredictable, also consumes a lot of precious time. Also, administrators in agriculture and decision-makers should be easily able to comprehend the forecasted results. A very confined and limited research has been performed in the field of soft computing based techniques for crop planning and crop resource harnessing. The scope of the current prediction systems is circumscribed to case studies and laboratories only. Clearly, these systems would prove impracticable in the field of agriculture since they cannot demonstrate the diversity and intricacy of crop production.

3. Related work

Fuzzy time series prediction is a prudent avenue in the areas where information is inexplicit, unclear and approximate. Also, fuzzy time series can tackle circumstances that neither provides the analysis of trends nor the visualization of patterns in time series. Profound research work has been accomplished on forecasting problems using this concept. Vikas [1] proposed different techniques for prediction of crop yields and used the artificial neural network to predict wheat yield. Adesh [2] did a comparative study of different techniques involving neural networks and fuzzy models. Askar [3] also tried to predict crop yield using time series models. Sachin [4,5] worked specifically on rice yield prediction using fuzzy time series model. Narendra [6] tried to predict wheat yield. Pankaj [7] used adaptive neuro-fuzzy systems for crop yield forecasting Wheat Yield Prediction. Fuzzy time series concepts and definitions were invented and presented by Song and Chissom. They also portrayed the concepts and notions of variant and invariant time series [8,9]. Initially, time series data of the University of Alabama was taken and enrollment forecasting was executed, and after some years they also [10] formulated an average autocorrelation function as a measure of dependency. Later, Chen [11,12] depicted simplified arithmetic operations instead of using max-min composition operations that were previously accustomed by Song & Chissom and then, arranged forecasted model using high order fuzzy time series. Singh [13] proposed fuzzy forecasting method, with a slight variation. Lee administered a fuzzy candlestick pattern to enhance forecasting outcomes [14]. Later, a multivariate heuristic model was designed and implemented to obtain highly intricate and complex matrix computations [15]. Research work was performed to ascertain the length of Intervals of fuzzy time series [16]. Event discretization function based forecasting models were put forth [17] and practiced to predict the average duration of stay of a patient [18]. Garg [19,20] developed a forecasting approach by administering the notion of OWA weights. This model proved to be an accomplishment as it downsized forecasting error to a certain extent. Afterwards, Garg [21,22] also put forward an optimized model based on genetic-fuzzy-OWA forecasting. Garg [23] proposed a new prediction model on time series data. Subsequently, the number of outpatient visits in the hospital was demonstrated by Garg [24].

As a matter of fact, the majority of these models were administered for prediction of all other problem domains except wheat production. Keeping this fact in mind, one can put forth a model to predict wheat production for India on the premise of historical time series wheat data. Real-time data of Pantnagar farm, G.B. Pant University of Agriculture & Technology, India

Table 1
Production values.

Year	Production
1981	2730
1982	2957
1983	2382
1984	2572
1985	2642
1986	2700
1987	2872
1988	3407
1989	2238
1990	2895
1991	3276
1992	1431
1993	2248
1994	2857
1995	2318
1996	2617
1997	2254
1998	2910
1999	3434
2000	2795

Table 2
Frequency distribution (5 interval).

Fuzzy sets	Lower	Upper	Frequency
F1	1400	1820	1
F2	1820	2240	1
F3	2240	2660	6
F4	2660	3080	9
F5	3080	3500	3

has been used by us. Then, the authors have applied their model to the aforesaid data. Subsequently, the authors have tried to provide a unique blend of Fuzzy logic and Regression technique to work on the data.

4. Proposed method

This section proposes a method for wheat production forecasting by using actual production as the universe of discourse and intervals based partitioning. We have provided another method for forecasting the value, which would be clearly explained in the lines to come. The forecasting process follows the following steps:

Step 1: First, clearly depict the universe of discourse U and partition U into equally length intervals.

One would specify the universe of discourse, i.e. the intervals within which all the given values of wheat production would lie. For defining the universe of discourse, the minimum value of production D_{min} and the maximum value of production D_{max} of given historical are first defined. Here, according to the data, 1431 is the minimum value and 3434 is the maximum value. Thus the Universe of discourse U is defined as $U = [D_{min}-D1, D_{max}+D2]$ where $D1$ and $D2$ are two positive numbers. Thus, in this case, the Universe of Discourse would be $[1400, 3500]$. The historical data is given year wise in Table 1.

Step 2: Now, depict fuzzy sets F_i as follows:

This research work showcases work at four different intervals to cover a wide range of partitions. The universe of discourse is divided into 5 equal intervals, 7 equal intervals, 9 equal intervals and 11 equal intervals that are shown in Tables 2, 4, 6 and 8 respectively. The intervals are further divided in proportion to the frequency of the data values that lie within each partition. Let's call this method as frequency-based partitioning, which is shown in Tables 3, 5, 7 and 9 for each interval.

(a) 5 equal intervals

Fuzzy sets are divided in frequency based partition
Similarly, 7, 9 and 11 intervals are also calculated,

(b) 7 equal intervals

Table 3
Frequency based partition (5 interval).

Fuzzy sets	Lower	Upper	New sets
F1	1400	1820	A1
F2	1820	2240	A2
F3	2240	2310	A3
	2310	2380	A4
	2380	2450	A5
	2450	2520	A6
	2520	2590	A7
	2590	2660	A8
F4	2660	2706.7	A9
	2706.7	2753.4	A10
	2753.4	2800.1	A11
	2800.1	2846.8	A12
	2846.8	2893.5	A13
	2893.5	2940.2	A14
	2940.2	2986.9	A15
	2986.9	3033.6	A16
	3033.6	3080.3	A17
F5	3080.3	3220.3	A18
	3220.3	3360.3	A19
	3360.3	3500	A20

In same fashion, first check the frequency of distribution in each value and then make a new table accordingly. Fuzzy sets are divided in frequency based partition

(c) 9 equal intervals

Fuzzy sets are divided in frequency based partition:

(d) 11 equal intervals

Fuzzy sets are divided in frequency based partition

STEP 3: Now, fuzzify the data using fuzzy Logic Relationships (FLR) mapping.

Once, the frequency based partitioning has been obtained, each of partition is denoted by F (I), where (I) signifies within which Intervals does the value lie. The increase in the value of “I” signifies increase in the value of the output. This nomenclature has been used to give a meaningful insight to the reader.

For example, while working on 7 intervals, each partition can be denoted by fuzzy intervals as shown below:

F1: VERY POOR YIELD
 F2: POOR YIELD
 F3: NOT SO GOOD YIELD
 F4: AVERAGE PRODUCTION
 F5: GOOD YIELD
 F6: VERY GOOD YIELD
 F7: EXCELLENT YIELD

Thus, increase in the suffix (I) is clearly associated with higher yield in the production of wheat and giving this nomenclature helps the reader to easily understand what each interval signifies instead of going deep into the mathematical jargon.

After this classification, fuzzy logic relationships (FLR) are established among the given set of values. FLR establish links among the fuzzified intervals of the data set. Using that, one can predict the forecast value by using a defuzzification algorithm which can be applied to these relationships. This can be explained through the given example:

Suppose, referring to Table 4;

In 1981, Yield = 2730 (belongs to F5)

In 1982, Yield = 2957 (belongs to F6)

In 1983, Yield = 2382 (belongs to F4)

In 1984, Yield = ? (Suppose this needs to be predicted, let F be the interval in which this value lies.)

A 1st order, Fuzzy Logic Relationship would look like:

$F6 \leftarrow F5$

$F4 \leftarrow F6$

Table 4

Frequency distribution (7 interval).

Fuzzy sets	Lower	Upper	Frequency
F1	1400	1700	1
F2	1700	2000	0
F3	2000	2300	3
F4	2300	2600	3
F5	2600	2900	8
F6	2900	3200	2
F7	3200	3500	3

Table 5

Frequency based partition (7 interval).

Fuzzy sets	Lower	Upper	New sets
F1	1400	1700	B1
F3	2000	2100	B2
	2100	2200	B3
	2200	2300	B4
F4	2300	2400	B5
	2400	2500	B6
	2500	2600	B7
F5	2600	2637.5	B8
	2637.5	2675	B9
	2675	2712.5	B10
	2712.5	2750	B11
	2750	2787.5	B12
	2787.5	2825	B13
	2825	2862.5	B14
	2862.5	2900	B15
F6	2900	3050	B16
	3050	3200	B17
F7	3200	3340	B18
	3340	3480	B19
	3480	3500	B20

Table 6

Frequency distribution (9 interval).

Fuzzy sets	Lower	Upper	Frequency
F1	1400	1633.33	1
F2	1633.33	1866.66	0
F3	1866.66	2099.99	0
F4	2099.99	2333.32	4
F5	2333.32	2566.65	1
F6	2566.65	2799.98	5
F7	2799.98	3033.31	5
F8	3033.31	3266.64	1
F9	3266.64	3500	3

And so on...

These logical relationships help to predict the value of a particular year using the fuzzified values of previous years by establishing a relation among them.

In similar fashion, a 2nd order Fuzzy Logic Relationship would look like:

$$F4 \leftarrow F6, F5$$

That means, F4 interval can be predicted using the previous two intervals F6 and F5.

Also, a 3rd Order fuzzy logic relationship would look like:

$$F \leftarrow F4, F6, F5$$

Here F is the predicted interval for yield in year 1984.

Now, a suitable defuzzification algorithm (discussed in step 5) can be applied on these values to predict value of the yield in year 1984, given that suitable equation is used to map the relationship of F with F4, F6 and F5, which correspond to the fuzzy intervals at years 1983, 1982 and 1981.

Table 7

Frequency based partition (9 interval).

Fuzzy sets	Lower	Upper	New fuzzy sets
F1	1400	1633.33	C1
F4	2099.99	2158.32	C2
	2158.32	2216.65	C3
	2216.65	2274.98	C4
	2274.98	2333.31	C5
F5	2333.31	2566.65	C6
F6	2566.65	2613.31	C7
	2613.31	2659.97	C8
	2659.97	2706.63	C9
	2706.63	2753.29	C10
F7	2753.29	2799.95	C11
	2799.95	2846.61	C12
	2846.61	2893.27	C13
	2893.27	2939.93	C14
F8	2939.93	2986.59	C15
	2986.59	3033.25	C16
	3033.25	3266.64	C17
	3266.64	3344.41	C18
F9	3344.41	3422.19	C19
	3422.19	3500	C20

Table 8

Frequency distribution (11 interval).

Fuzzy sets	Lower	Upper	Frequency
F1	1400	1590.9	1
F2	1590.9	1781.8	0
F3	1781.8	1972.7	0
F4	1972.7	2163.6	0
F5	2163.6	2354.5	4
F6	2354.5	2545.4	1
F7	2545.4	2736.3	5
F8	2736.3	2927.2	5
F9	2927.2	3118.1	1
F10	3118.1	3309	2
F11	3309	3500	1

Step 4: Take the average of the mid points of the fuzzified intervals present on the RHS of the fuzzy logic relationship (FLR).

For example, in the 2nd order FLR,

$$F4 \leftarrow F6, F5$$

If A is the midpoint of interval F6 and B is the midpoint of interval F5, then calculate:

$$C = (A + B)/2$$

where C is the midpoint of the fuzzy interval F4 here

Similarly, for 3rd order FLR,

$$F \leftarrow F4, F6, F5$$

If A is the midpoint of interval F6, B is the midpoint of interval F5 and C is the midpoint of F4, then calculate:

$$D = (A + B + C)/3,$$

where, D is the average fuzzified value for the year 1984. Now this would be used as a variable to linear regression model, for complete defuzzification. Using this obtained result; one can easily compute the predicted value.

Just as observed in the above equations, one can compute the midpoints of the fuzzy intervals and divide their sum by the degree of the fuzzy logical relationships that exist among them. Similarly, one can compute this result for the next value, until all the values have been parsed. So the general formula looks like:

$$\text{Average Fuzzified Value} = (\text{Sum of previous } n \text{ fuzzy values})/(n)$$

where, n denotes FLR degree (which is 2 or 3 in our model).

Using above formula, following Tables 10 and 11 are computed for 5, 7, 9 and 11 intervals in second order and third order FLR.

Table 9
Frequency based partition (11 interval).

Fuzzy sets	Lower	Upper	New fuzzy sets
F1	1400	1590.9	D1
F5	2163.6	2211.325	D2
	2211.325	2259.05	D3
	2259.05	2306.775	D4
	2306.775	2354.5	D5
F6	2354.5	2545.4	D6
F7	2545.4	2583.58	D7
	2583.58	2621.76	D8
	2621.76	2659.94	D9
	2659.94	2698.12	D10
	2698.12	2736.3	D11
	2736.3	2774.48	D12
F8	2774.48	2812.66	D13
	2812.66	2850.84	D14
	2850.84	2889.02	D15
	2889.02	2927.2	D16
	2927.2	3118.1	D17
F9	3118.1	3213.55	D18
F10	3213.55	3309	D19
	3309	3500	D20
F11	3309	3500	D20

Table 10
FLR second degree and AVG for all four intervals.

YEAR	YIELD	FLR 5 intervals	AVG 1	FLR 7 intervals	AVG 2	FLR 9 intervals	AVG 3	FLR 11 intervals	AVG 4
1981	2730	–	–	–	–	–	–	–	–
1982	2957	–	–	–	–	–	–	–	–
1983	2382	A12<-A11,A10	2846.8	B5<-B16,B11	2853.13	C6<-C15,C10	2846.61	D6<-D17,D11	2869.93
1984	2572	A13<-A12,A11	2689.275	B7<-B5,B16	2662.5	C7<-C6,C15	2706.62	D7<-D6,D17	2736.3
1985	2642	A14<-A13,A12	2485	B9<-B7,B5	2450	C8<-C7,C6	2519.98	D9<-D7,D6	2507.22
1986	2700	A15<-A14,A13	2619.175	B10<-B9,B7	2603.13	C9<-C8,C7	2613.31	D11<-D9,D7	2602.67
1987	2872	A16<-A15,A14	2706.7	B15<-B10,B9	2675	C13<-C9,C8	2659.97	D15<-D11,D9	2679.03
1988	3407	A17<-A16,A15	2800.1	B19<-B15,B10	2787.5	C19<-C13,C9	2776.62	D20<-D15,D11	2793.57
1989	2238	A18<-A17,A16	3150.15	B4<-B19,B15	3145.63	C4<-C19,C13	3126.62	D3<-D20,D15	3137.215
1990	2895	A19<-A18,A17	2730.075	B15<-B4,B19	2830	C14<-C4,C19	2814.5575	D16<-D3,D20	2819.84375
1991	3276	A20<-A19,A18	2473.425	B18<-B15,B4	2565.63	C18<-C14,C4	2581.2075	D19<-D16,D3	2571.64875
1992	1431	A21<-A20,A19	3103.575	B1<-B18,B15	3075.63	C1<-C18,C14	3111.0625	D1<-D19,D16	3084.6925
1993	2248	A22<-A21,A20	2450.15	B4<-B1,B18	2410	C4<-C1,C18	2411.095	D3<-D1,D19	2378.3625
1994	2857	A23<-A22,A21	1942.5	B14<-B4,B1	1900	C13<-C4,C1	1881.24	D15<-D3,D1	1865.31875
1995	2318	A24<-A23,A22	2572.575	B5<-B14,B4	2546.88	C5<-C13,C4	2557.8775	D5<-D15,D3	2552.55875
1996	2617	A25<-A24,A23	2607.575	B8<-B5,B14	2596.88	C8<-C5,C13	2587.0425	D8<-D5,D15	2600.28375
1997	2254	A26<-A25,A24	2485	B4<-B8,B5	2484.38	C4<-C8,C5	2470.3925	D4<-D8,D5	2466.65375
1998	2910	A27<-A26,A25	2450	B16<-B4,B8	2434.38	C14<-C4,C8	2441.2275	D16<-D4,D8	2442.79125
1999	3434	A28<-A27,A26	2595.925	B19<-B16,B4	2612.5	C20<-C14,C4	2581.2075	D20<-D16,D4	2595.51125
2000	2795	A29<-A28,A27	3173.5	B13<-B19,B16	3192.5	C11<-C20,C14	3188.845	D13<-D20,D16	3156.305

Tables 10 and 11 show a consolidated view of FLR mapping of second and third degree respectively. AVG column explains the average of the mid-points of fuzzy intervals taken according to FLR mapping.

Step 5: Now, Let's introduce the concept of regression analysis to defuzzify the fuzzified values, which were calculated in Step 4. Plot a graph wherein,

X-axis → denotes the respective years (taken in form of 1, 2, 3, 4.....so on).

Y-axis → denotes the values (average fuzzified values, which are denoted by AVG column in subsequent tables) obtained in Step 4. Once these points are plotted, choose a Line of Best Fit that averages through all the points in the graph. Then, compute the equation of this line, which is either linear or a polynomial of degree 2, 3 or 4. Following tables have used two parameters to compare the results, these are as follows:

(a) Average Forecasting Error Rate (AFER)

$$AFER = \left(\sum_{i=1}^n (|A_i - F_i|/A_i) \right) / n * 100\%$$

(b) Mean Square Error (MSE).

$$MSE = \left(\sum_{i=1}^n (A_i - F_i)^2 \right) / n$$

Table 11

FLR third degree and AVG for all four intervals.

YEAR	YIELD	FLR 5 intervals	AVG 5 intervals	FLR 7 intervals	AVG 7 intervals	FLR 9 intervals	AVG 9 intervals	FLR 11 intervals	AVG 11 intervals
1981	2730	-	-	-	-	-	-	-	-
1982	2957	-	-	-	-	-	-	-	-
1983	2382	-	-	-	-	-	-	-	-
1984	2572	A7 <-A5,A15,A10	2702.86	B7 <-B5,B16,B11	2685.4	C7 <-C6,C15,C10	2714.4	D7 <-D6,D17,D11	2729.9
1985	2642	A9 <-A7,A5,A15	2644.516	B9 <-B7,B5,B16	2625	C8 <-C7,C6,C15	2667.74	D9 <-D7,D6,D17	2679
1986	2700	A10 <-A9,A7,A5	2551.116	B10 <-B9,B7,B5	2518.8	C9 <-C8,C7,C6	2558.866667	D11 <-D9,D7,D6	2551.8
1987	2872	A13 <-A10,A9,A7	2656.1333	B15 <-B10,B9,B7	2633.3	C13 <-C9,C8,C7	2636.64	D15 <-D11,D9,D7	2640.9
1988	3407	A20 <-A13,A10,A9	2761.1833	B19 <-B15,B10,B9	2743.8	F19 <-F13,F9,F8	2729.96	D20 <-D15,D11,D9	2742.7
1989	2238	A2 <-A20,A13,A10	3010.1166	B4 <-B19,B15,B10	2995	C4 <-C19,C13,C9	2978.846667	D3 <-D20,D15,D11	2997.2
1990	2895	A14 <-A2,A20,A13	2776.7666	B15 <-B4,B19,B15	2847.1	C14 <-C4,C19,C13	2833.018333	D16 <-D3,D20,D15	2836.5
1991	3276	A19 <-A14,A2,A20	2792.3333	B18 <-B15,B4,B19	2847.1	C18 <-C14,C4,C19	2848.571667	D19 <-D16,D3,D20	2849.3
1992	1431	A1 <-A19,A14,A2	2745.7166	B1 <-B18,B15,B4	2800.4	C1 <-C18,C14,C4	2822.646667	D1 <-D19,D16,D3	2801.5
1993	2248	A3 <-A1,A19,A14	2605.7166	B4 <-B1,B18,B15	2567.1	C4 <-C1,C18,C14	2579.596667	D3 <-D1,D19,D16	2554.9
1994	2857	A13 <-A3,A1,A19	2391.7666	B14 <-B4,B1,B18	2356.7	C13 <-C4,C1,C18	2356.001667	D15 <-D3,D1,D19	2330.6
1995	2318	A4 <-A13,A3,A1	2251.7166	B5 <-B14,B4,B1	2214.6	C5 <-C13,C4,C1	2210.806667	D5 <-D15,D3,D1	2200.2
1996	2617	A8 <-A4,A13,A3	2496.7166	B8 <-B5,B14,B4	2481.3	C8 <-C5,C13,C4	2473.3	D8 <-D5,D15,D3	2478.6
1997	2254	A3 <-A8,A4,A13	2613.3833	B4 <-B8,B5,B14	2604.2	C4 <-C8,C5,C13	2603.575	D4 <-D8,D5,D15	2601.1
1998	2910	A14 <-A3,A8,A4	2415	B16 <-B4,B8,B5	2406.3	C14 <-C4,C8,C5	2395.533333	D16 <-D4,D8,D5	2405.4
1999	3434	A20 <-A14,A3,A8	2605.6166	B19 <-B16,B4,B8	2614.6	C20 <-C14,C4,C8	2599.685	D20 <-D16,D4,D8	2597.9
2000	2795	A11 <-A20,A14,A3	2874	B13 <-B19,B16,B4	2878.3	C11 <-C20,C14,C4	2874.501667	D13 <-D20,D16,D4	2865.2

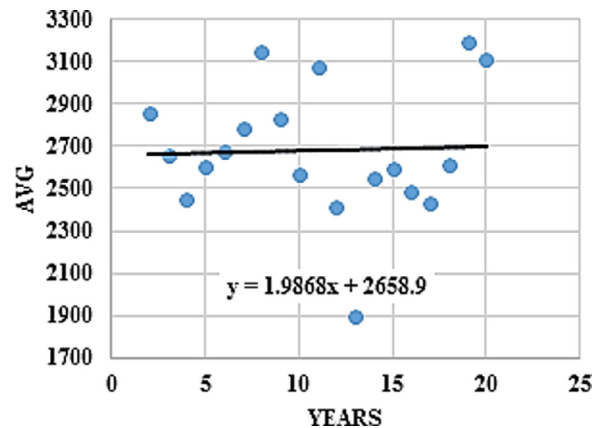


Fig. 1. (5 Intervals) refers Table 10.

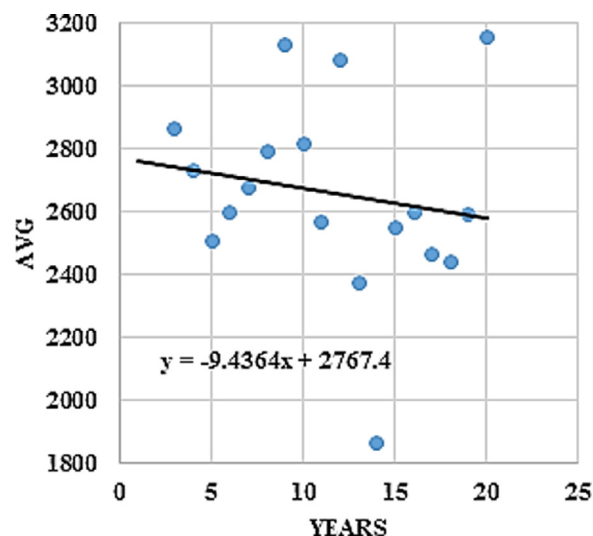


Fig. 2. (7 Intervals) refers Table 10.

Where, A_i denotes the real time production value and

F_i denotes the predicted value of year i .

I. Linear

$Y = Mx + C$, where Y will give the predicted value for the year and X is input to this equation in the form of (1, 2, 3, 4...so on, which corresponds to years 1981, 1982, 1983, 1984...so on). By using Figs. 1–4, forecast values for each year can be computed and then the Mean Square Error (MSE) and Average forecasting error rate (AFER) is calculated corresponding to Table 12.

It can be seen that in Figs. 1–4, the plot of 5, 7, 9 and 11 intervals has been done for FLR 2nd-degree relationship. The line came out as a polynomial degree 1 equation and it is called as the line of best fit, which passes through most of the points in the best way possible. Table 12 represents consolidated forecast values of all four intervals.

It can be noted that the lowest MSE has been obtained for interval 7. So the best choice of interval for predictive modeling, in this case, turns out to be 7th interval partitioning.

As it can be seen that Table 12 has been constructed using the forecast values obtained from equations generated in Figs. 1–4. This procedure has been done for second-degree FLR relations. A similar procedure can be done for third-degree FLR relations. After applying the same algorithm, it can repeat for FLR third degree.

II. Polynomial degree 2

$Y = Ax^2 + Bx + C$, where Y will give the predicted value for the year and X is input to this equation in the form of (1, 2, 3, 4...so on, which corresponds to years 1981, 1982, 1983, 1984...so on). Using Figs. 5–8, the forecast values for each year

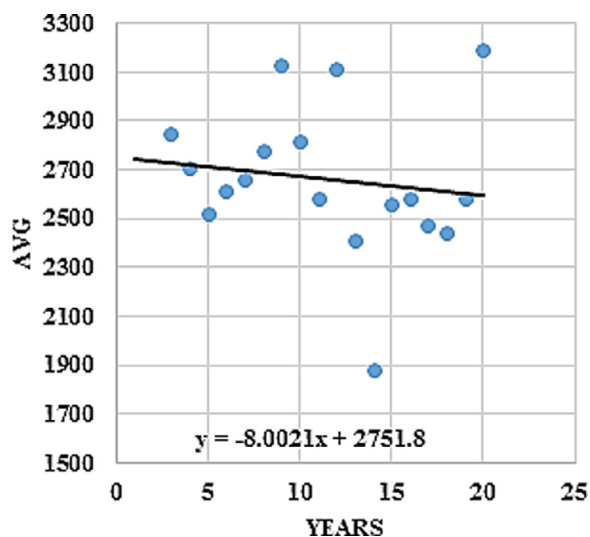


Fig. 3. (9 Intervals) refers Table 10.

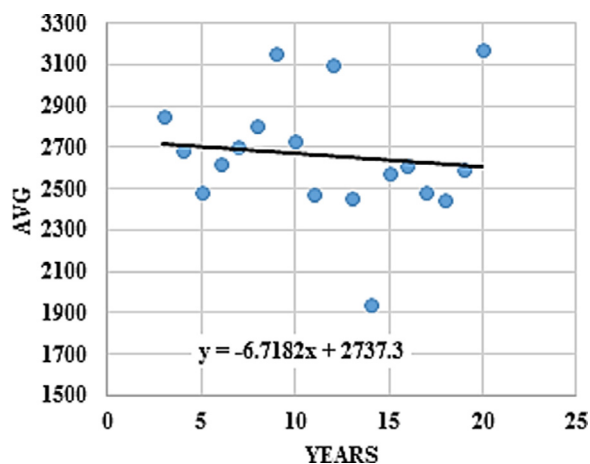


Fig. 4. (11 Intervals) refers Table 10.

Table 12

MSE and AFER using polynomial degree 1 and FLR second degree.

5 Intervals	7 Intervals	9 Intervals	11 Intervals
2723.8636	2662.8736	2375.7958	2748.527
2717.1454	2664.8604	2727.7937	2739.091
2710.4272	2666.8472	2719.7916	2729.654
2703.709	2668.834	2711.7895	2720.218
2696.9908	2670.8208	2703.7874	2710.782
2690.2726	2672.8076	2695.7853	2701.345
2683.5544	2674.7944	2687.7832	2691.909
2676.8362	2676.7812	2679.7811	2682.472
2670.118	2678.768	2671.779	2673.036
2663.3998	2680.7548	2663.7769	2663.6
2656.6816	2682.7416	2655.7748	2654.163
2649.9634	2684.7284	2647.7727	2644.727
2643.2452	2686.7152	2639.7706	2635.29
2636.527	2688.702	2631.7685	2625.854
2629.8088	2690.6888	2623.7664	2616.418
2623.0906	2692.6756	2615.7643	2606.981
2616.3724	2694.6624	2607.7622	2597.545
2609.6542	2696.6492	2599.7601	2588.108
2602.936	2698.636	2591.758	2578.672
MSE = 216,554	MSE = 214,089	MSE = 217,285	MSE = 218,231
AFER = 15.053	AFER = 14.98	AFER = 0.15	AFER = 0.15

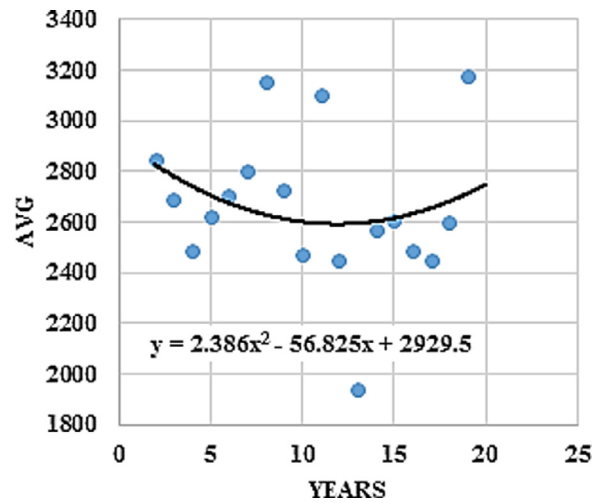


Fig. 5. (5 Intervals) refers Table 10.

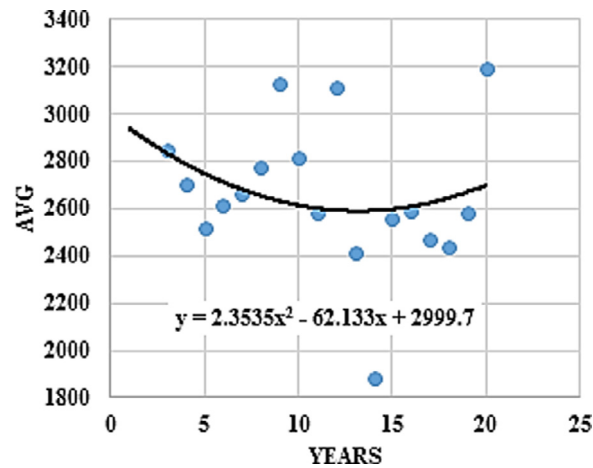


Fig. 6. (7 Intervals) refers Table 10.

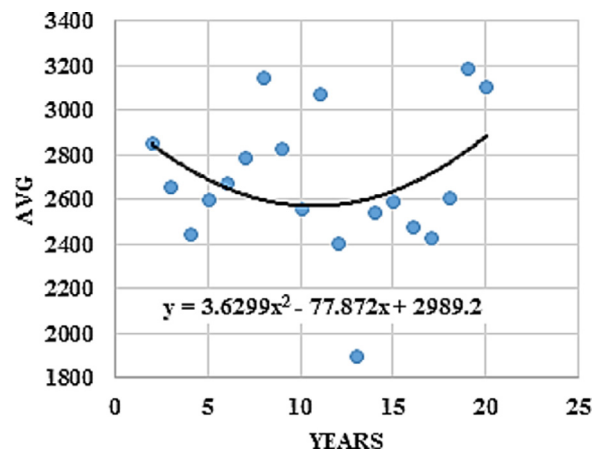


Fig. 7. (9 Intervals) refers Table 10.

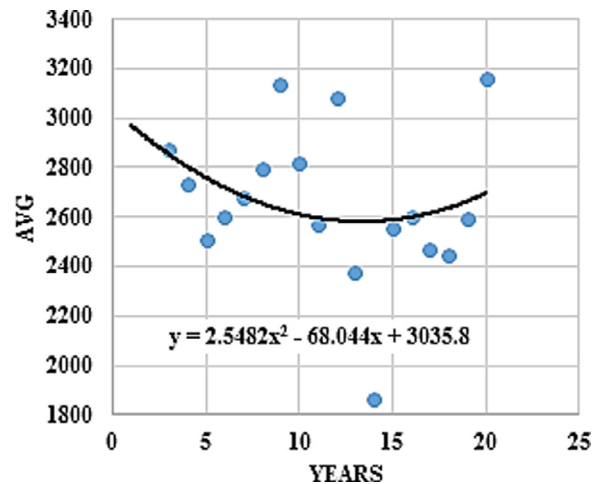


Fig. 8. (11 Intervals) refers Table 10.

Table 13

MSE and AFER using polynomial degree 2 and FLR second degree.

5 Intervals	7 Intervals	9 Intervals	11 Intervals
2825.394	2847.9756	2884.848	2909.905
2780.499	2788.2531	2834.4825	2854.602
2740.376	2735.7904	2788.824	2804.395
2705.025	2690.5875	2747.8725	2759.285
2674.446	2652.6444	2711.628	2719.271
2648.639	2621.9611	2680.0905	2684.354
2627.604	2598.5376	2653.26	2654.533
2611.341	2582.3739	2631.1365	2629.808
2599.85	2573.47	2613.72	2610.18
2593.131	2571.8259	2601.0105	2595.648
2591.184	2577.4416	2593.008	2586.213
2594.009	2590.3171	2589.7125	2581.874
2601.606	2610.4524	2591.124	2582.631
2613.975	2637.8475	2597.2425	2588.485
2631.116	2672.5024	2608.068	2599.435
2653.029	2714.4171	2623.6005	2615.482
2679.714	2763.5916	2643.84	2636.625
2711.171	2820.0259	2668.7865	2662.864
2747.4	2883.72	2698.44	2694.2
MSE = 206,054	MSE = 203,066	MSE = 209,507	MSE = 210,399
AFER = 14.5	AFER = 14.60	AFER = 0.14	AFER = 0.148

are computed. Then, the Mean Square Error (MSE) and Average forecasting error rate (AFER) was calculated corresponding to each figure in Table 13.

It can be seen that in Figs. 5–8, a plot of 5, 7, 9 and 11 intervals has been done for FLR 2nd-degree relationship. The line came out as a polynomial degree 2 equation and it is called as the line of best fit. Table 13 represents consolidated forecast values of all four intervals. It can be noted that the lowest MSE has been obtained in interval 7. So, the best choice of interval for predictive modeling, in this case, is 7th interval partitioning.

It is clear from the figures above that the equation generated is a second-degree equation. Using these equations, the forecast values have been obtained in Table 13 for all the intervals 5, 7, 9 and 11. In a similar manner, forecast values of FLR third degree relations can be obtained.

III. Polynomial of degree 3

$Y = Ax^3 + Bx^2 + Cx + D$, where Y will give the predicted value for the year and X is input to this equation in the form of (1, 2, 3, 4....so on, which corresponds to years 1981, 1982, 1983, 1984....so on). Using Figs. 9–12, one can compute the forecast values for each year and then one can calculate the Mean Square Error (MSE) and Average forecasting error rate (AFER) corresponding to Table 14.

It can be seen that in Figs. 9–12, the plot of 5, 7, 9 and 11 intervals has been done for FLR 2nd-degree relationship. The line came out as a polynomial degree 3 equation and it is called as the line of best fit. Table 14 represents consolidated

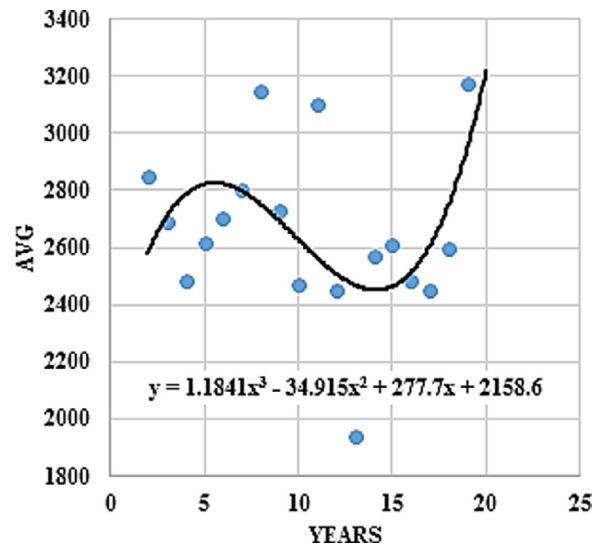


Fig. 9. (5 Intervals) refers Table 10.

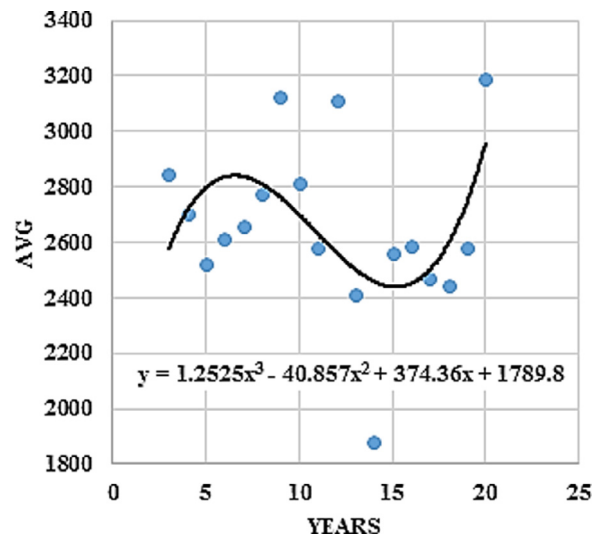


Fig. 10. (7 Intervals) refers Table 10.

forecast values of all four intervals and it can be noted that the lowest MSE has been obtained in interval 7. So the best choice of interval for predictive modeling, in this case, turns out to be 7th interval partitioning.

Table 14 clearly shows the forecast values obtained using third-degree polynomial equations generated in Figs. 9–12 for FLR second degree relations. Applying the same algorithm to FLR third degree relations, forecast values can be obtained in similar fashion.

IV. Polynomial of degree 4

$Y = Ax^4 + Bx^3 + Cx^2 + Dx + E$, Where, Y will give the predicted value for the year and X is input to this equation in the form of (1, 2, 3, 4....so on, which corresponds to years 1981, 1982, 1983, and 1984....so on). Using Figs. 13–16, one can compute the forecasted values for each year and then one can calculate the Mean Square Error (MSE) and Average forecasting error rate (AFER) corresponding to Table 15.

It can be seen that in Figs. 13–16, the plot of 5, 7, 9 and 11 intervals has been done for FLR 2nd-degree relationship. The line came out as a polynomial degree 4 equation and it is called as the line of best fit, which passes through most of the points in the best way possible. Table 15 represents consolidated forecast values of all four intervals and it can be noted that the lowest MSE has been obtained in interval 5. So the best choice of interval for predictive modeling, in this case, turns out to be 5th interval partitioning.

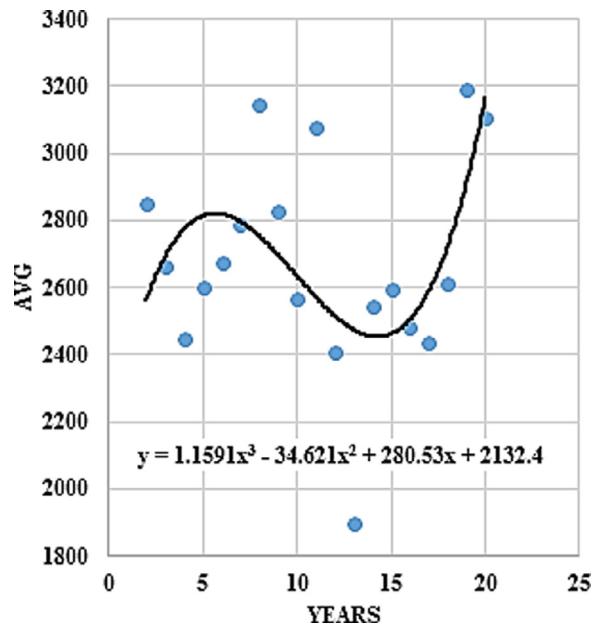


Fig. 11. (9 Intervals) refers Table 10.

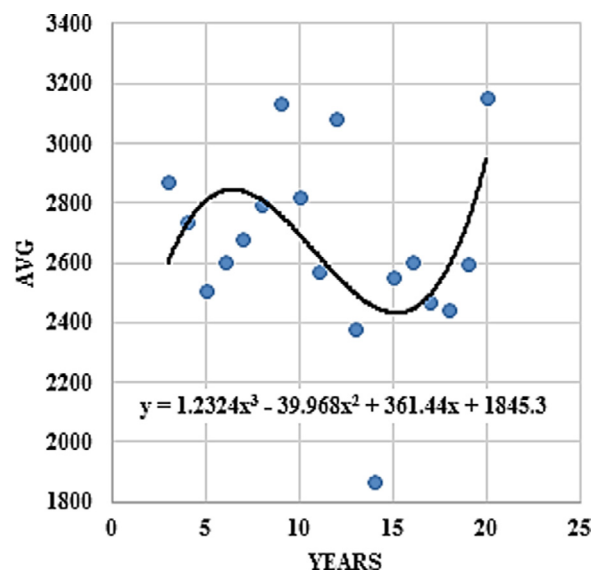


Fig. 12. (11 Intervals) refers Table 10.

Table 15 represents forecast values obtained using polynomial degree 4 equations plotted in Figs. 13–16. Applying the same algorithm, another table can be obtained that will represent forecast values of FLR third degree relations.

Hence, by using a concept, which is a blend of both fuzzy time series and regression analysis, we have put forth a method to predict wheat yield and it can be seen that 7th interval partitioning is a clear winner to give best predictions on data set with fewer errors. This has been clearly elucidated using graph visualization.

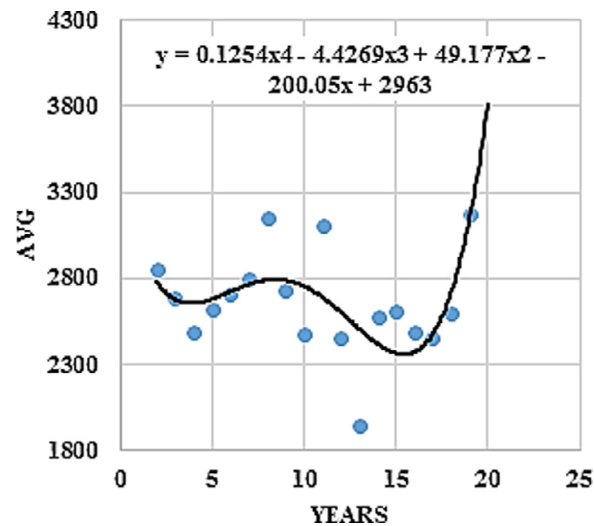
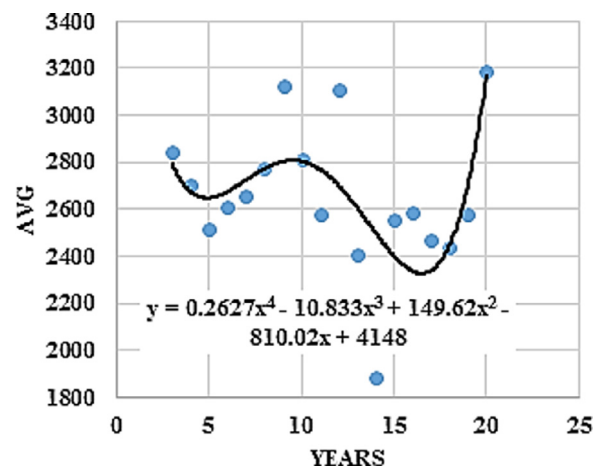
5. Results and discussion

The MSE and AFER as calculated above in the tables have been analyzed. The authors have worked on a second-order fuzzy logic relationship and third order fuzzy logic relationship (FLR). Among each degree, we have further worked on four

Table 14

MSE and AFER using polynomial degree 3 and FLR second degree.

5 Interval	7 Interval	9 Intervals	11 Intervals
2583.8128	2564.2488	–	–
2709.4357	2693.6967	2578.9845	2603.183
2786.5424	2774.7664	2713.688	2730.446
2822.2375	2814.4125	2796.7375	2807.35
2823.6256	2819.5896	2835.648	2841.29
2797.8113	2797.2523	2837.9345	2839.661
2751.8992	2754.3552	2811.112	2809.857
2692.9939	2697.8529	2762.6955	2759.272
2628.2	2634.7	2700.2	2695.3
2564.6221	2571.8511	2631.1405	2625.336
2509.3648	2516.2608	2563.032	2556.775
2469.5327	2474.8837	2503.3895	2497.011
2452.2304	2454.6744	2459.728	2453.438
2464.5625	2462.5875	2439.5625	2433.45
2513.6336	2505.5776	2450.408	2444.442
2606.5483	2590.5993	2499.7795	2493.809
2750.4112	2724.6072	2595.192	2588.945
2952.3269	2914.5559	2744.1605	2737.244
3219.4	3167.4	2954.2	2946.1
MSE = 185,738	MSE = 185,460	MSE = 188,576	MSE = 189,897
AFER = 14.693	AFER = 14.645	AFER = 0.1426	AFER = 0.143

**Fig. 13.** (5 Intervals) refers Table 10.**Fig. 14.** (7 Intervals) refers Table 10.

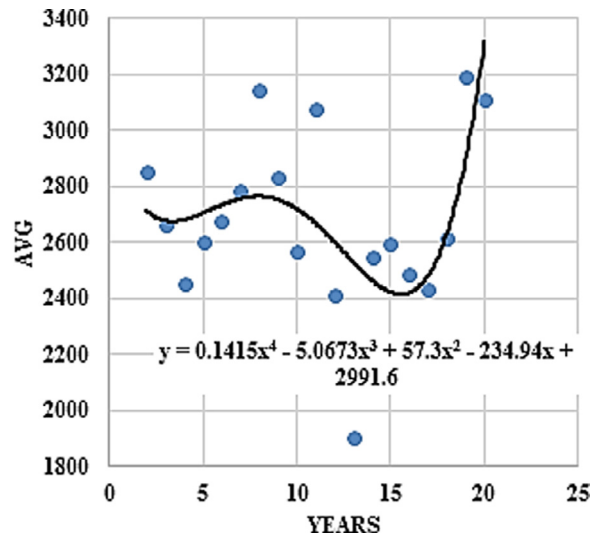


Fig. 15. (9 Intervals) refers Table 10.

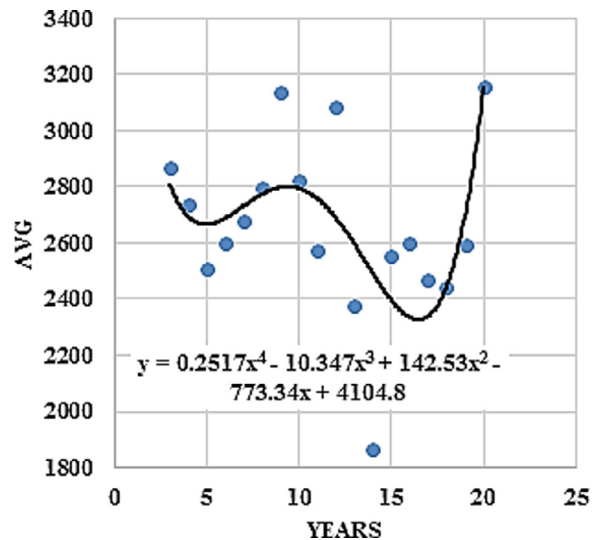


Fig. 16. (11 Intervals) refers Table 10.

different intervals such as 5th, 7th, 9th and 11th intervals. To reduce the complexity of our research and present it in an easier to understand fashion, a comparison is done among all the results in the form of easy to understand bar graphs.

As it can be seen, Fig. 17 shows the comparison of Mean Square Error (MSE) among all polynomial degrees in both second and third-degree fuzzy logic relationship (FLR) in the entire 5th interval. It can be inferred from this figure that the lowest MSE is achieved by the cubic equation in second-degree FLR.

In Fig. 18, we have shown the comparison of Mean Square Error (MSE) among all polynomial degrees in both second and third-degree fuzzy logic relationship (FLR) in the entire 7th interval. It can be inferred from this figure that the lowest MSE is achieved by the cubic equation in third-degree FLR.

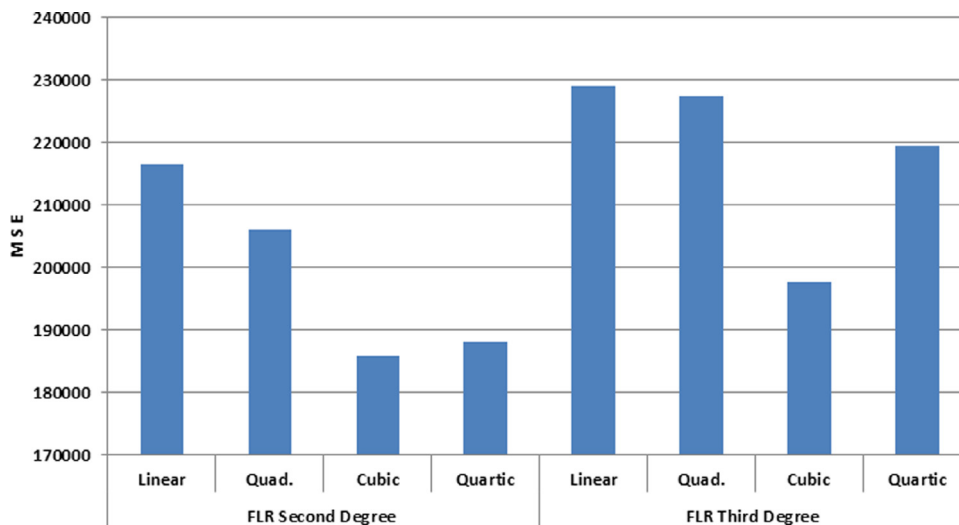
In Fig. 19, the comparison of Mean Square Error (MSE) was shown among all polynomial degrees in both second and third-degree fuzzy logic relationship (FLR) in the entire 9th interval. It can be inferred from this figure that the lowest MSE is achieved by the cubic equation in second-degree FLR.

In Fig. 20, the comparison of Mean Square Error (MSE) is shown among all polynomial degrees in both second and third-degree fuzzy logic relationship (FLR) in the entire 11th interval. It can be inferred from this figure that the lowest MSE is achieved by the cubic equation in second-degree FLR. It can be finally concluded that cubic equation gives the lowest possible MSE no matter which interval one is working in.

Table 15

MSE and AFER using polynomial degree 4 and FLR second degree.

5 Intervals	7 Intervals	9 Intervals	11 Intervals
2726.1992	2712.6456	–	–
2696.0741	2677.1244	2793.308	2808.569
2698.4128	2680.5568	2675.779	2694.147
2717.1875	2704.425	2648.463	2665.288
2739.38	2733.6072	2674.731	2687.091
2754.9817	2756.3776	2724.264	2730.701
2756.9936	2764.4064	2773.043	2773.299
2741.4263	2752.7598	2803.358	2798.111
2707.3	2719.9	2803.8	2794.4
2656.6445	2667.6852	2769.268	2757.473
2594.4992	2601.3696	2700.963	2688.675
2528.9131	2529.6034	2606.394	2595.395
2470.9448	2464.4328	2499.371	2491.059
2434.6625	2421.3	2400.013	2395.138
2437.144	2419.0432	2334.739	2333.139
2498.4767	2479.8966	2336.278	2336.615
2641.7576	2629.4904	2443.659	2443.155
2893.0933	2896.8508	2702.22	2696.393
3281.6	3314.4	3163.6	3146
MSE = 188,015.25	MSE = 189,297	MSE = 224,457	MSE = 223,708
AFER = 14.404	AFER = 14.36	AFER = 0.15	AFER = 0.15

**Fig. 17.** Comparison among all degrees in 5 intervals.

To decide the lowest MSE among all the scenarios that were considered, the lowest MSE was chosen each from Figs. 17–20 and was compared in Fig. 21. It was found that the lowest MSE was achieved by the cubic equation of third-degree FLR in the 7th interval.

Hence, after comparing the results, FLR 3rd degree, 7 Interval polynomial third degree, has the best result with MSE (Mean Square Error): 180,985.463.

Fig. 17 compares both FLR second degree and third-degree relations. Comparison is done among all polynomial degrees from one to four as explained in graphical figures as well.

Fig. 18 compares both FLR second degree and third-degree relations. Lowest MSE is achieved by cubic polynomial and FLR third degree. Comparison is done among all polynomial degrees from one to four as explained in graphical figures as well.

Similarly, all comparisons are done in Fig. 19 and it compares both FLR second degree and third-degree relations. Lowest MSE is achieved by the cubic polynomial and FLR second degree. Comparison is done among all polynomial degrees from one to four as explained in graphical figures as well.

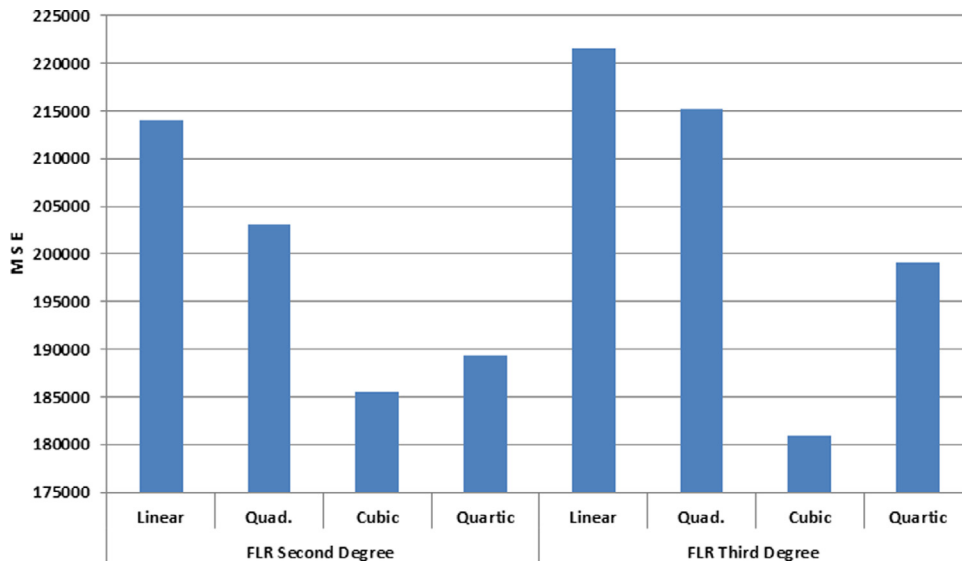


Fig. 18. Comparison among all degrees in 7 intervals.

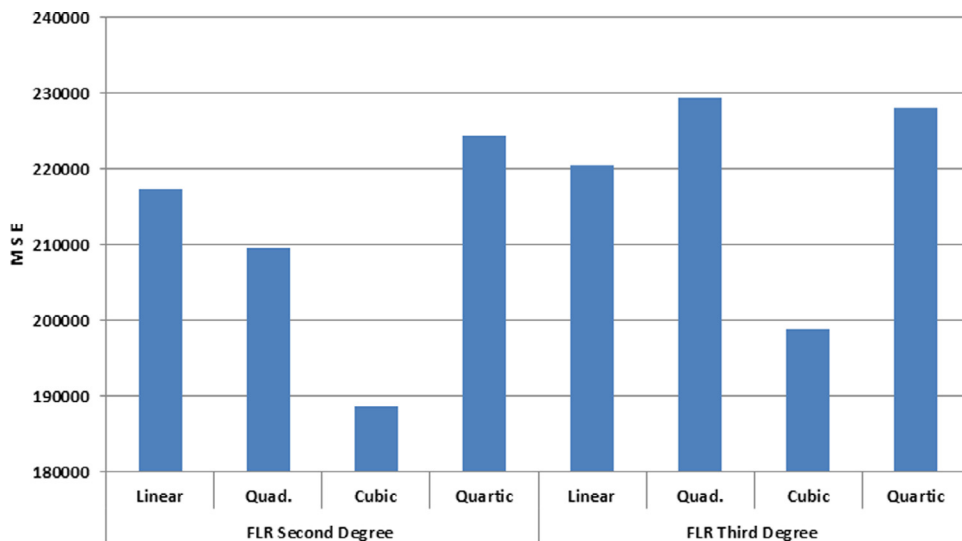


Fig. 19. Comparison among all degrees in 9 intervals.

In Fig. 20 also, the similar comparison is done and again it can be noticed that cubic polynomial equation shows the least MSE and thus the best results among those compared in this figure. Final Comparison is done in Fig. 21 among the best outcomes from each interval in order to show which case is best among all 32 cases taken into consideration.

6. Robustness

The robustness of the algorithm proposed by this paper has been proved by randomly increasing the production values of any given year and then the mean square error generated after applying this novel algorithm has been calculated.

As it can be seen in Table 16, the production values of four random years as 1992, 1994, 1996 and 1998 are increased by 5% and the MSE came out to be 188,224.88, which is just 4% increase from our actual MSE. The difference between the actual MSE and increased MSE is small enough and this small enough difference proves that even if the data set changes, this algorithm still remains applicable and gives stable MSE, which proves the strength of this algorithm. Similarly, in another case, the MSE is decreased by 5% in the same four years and MSE came out to be 190,034.73, which is just 5% increase from

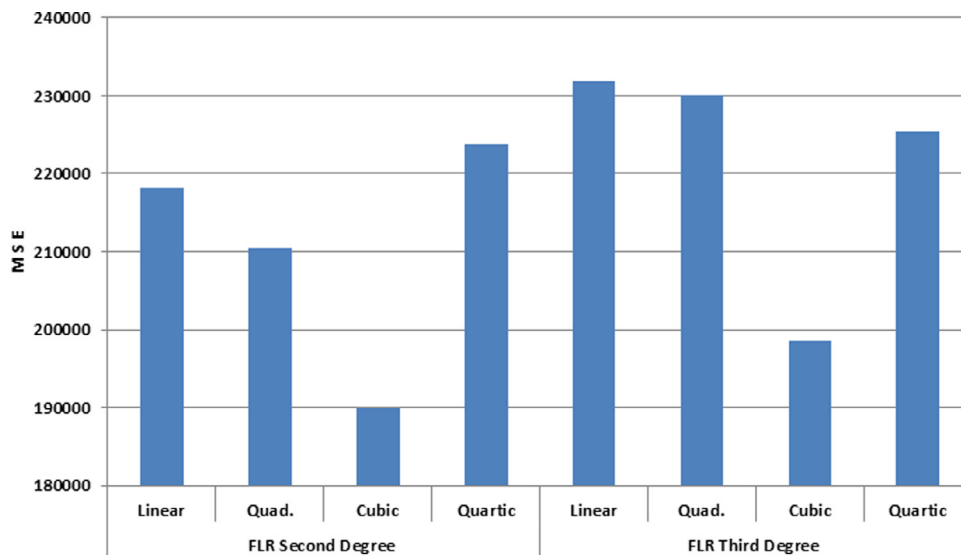


Fig. 20. Comparison among all degrees in 11 intervals.

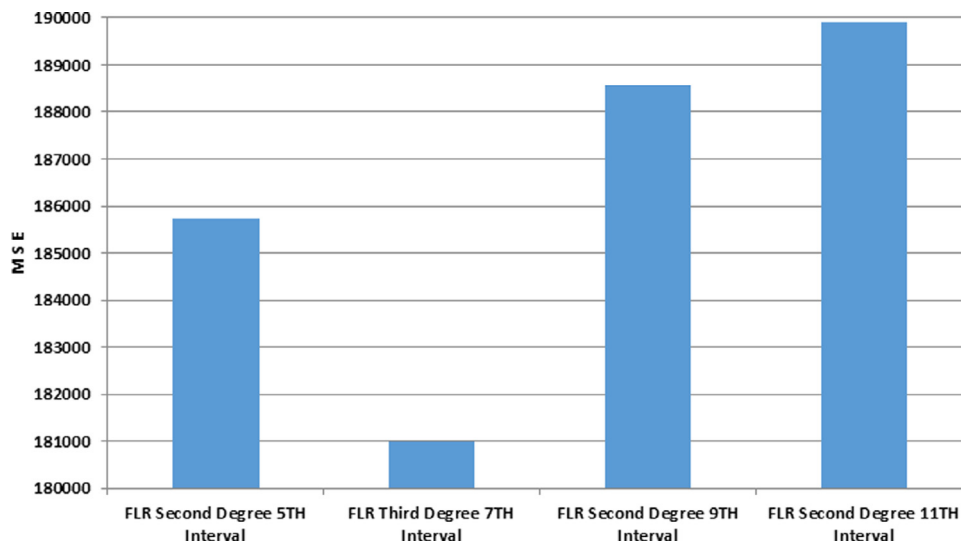


Fig. 21. Comparison among all intervals' best outcomes.

actual MSE and is very close to the actual MSE. So it can be concluded that the proposed algorithm is robust and can be used to do predictions in wide domains of data and not just the rice data production set.

7. Conclusion and future scope

It has been noticed that the proposed method is optimal and produces a high precision by having a small mean square error and average forecasting error rate. The developed fuzzy approach can be viewed as an inerrant and efficient way to assess, evaluate and estimate wheat production. Considering the future scope of this work, the proposed model can be extended to deal with multidimensional time series data and optimized with more advanced algorithms. We also think that various other degrees of Fuzzy Logical Relationship can be applied to the data, to see how it produces results for very high order FLR's. Another concept that we would like to work upon is to choose a different and more efficient method to partition the Universe of Discourse. In future, a new and improved Partitioning algorithm can be designed that on executing,

Table 16
Increased and decreased values.

Year	Production	Decreased 5%	Increased 5%
1981	2730	2730	2730
1982	2957	2957	2957
1983	2382	2382	2382
1984	2572	2572	2572
1985	2642	2642	2642
1986	2700	2700	2700
1987	2872	2872	2872
1988	3407	3407	3407
1989	2238	2238	2238
1990	2895	2895	2895
1991	3276	3276	3276
1992	1431	1359	1502
1993	2248	2248	2248
1994	2857	2714	2999
1995	2318	2318	2318
1996	2617	2486	2747
1997	2254	2254	2254
1998	2910	2764	3055
1999	3434	3434	3434
2000	2795	2795	2795

produces the least possible error in comparison to the other partition methods. For this, we would like to study different mathematical models proposed by mathematicians in detail, so that some efficient methods among those could be filtered.

References

- [1] Lamba V, Dhaka VS. Wheat yield prediction using artificial neural network and crop prediction techniques. *Int J Res Appl Sci Eng Technol* 2014;2:330–41.
- [2] Pandey AK, Sinha AK, Srivastava VK. A comparative study of neural-network & fuzzy time series forecasting techniques – case study: wheat production. *Int J Comput Sci Network Secur Forecasting* 2008;8:382–7.
- [3] Choudhury A, Jones J. Crop yield prediction using time series models. *J Econ Econ Educ Res* 2014;15:53–68.
- [4] Kumar S, Kumar N. A novel method for rice production forecasting using fuzzy time series. *Int J Comput Sci Issues* 2012;9:455–9.
- [5] Kumar S, Kumar N. Two factor fuzzy time series model for rice forecasting. *Int J Comput Math Sci* 2015;4:56–61.
- [6] Kumar N, Ahuja S, Kumar V, Kumar A. Fuzzy time series forecasting of wheat production. *Int J Comput Sci Eng* 2010;2:635–40.
- [7] Kumar P. Crop yield forecasting by adaptive neuro fuzzy inference system. *Math Theory Model* 2011;1:1–7.
- [8] Song Q, Chissom BS. Fuzzy time series and its models. *Fuzzy Sets Syst* 1993;54:269–77.
- [9] Song Q, Chissom BS. Forecasting enrolments with fuzzy time series: part II. *Fuzzy Sets Syst* 1994;62:1–8.
- [10] Song Q. A note on fuzzy time series model selection with sample autocorrelation functions. *Int J Cybern Syst* 2003;34:93–107.
- [11] Chen SM. Forecasting enrolments based on fuzzy time series. *Fuzzy Sets Syst* 1996;81:311–19.
- [12] Chen SM. Forecasting enrolments based on high order fuzzy time series. *Int J Cybern Syst* 2010;33:1–16.
- [13] Singh SR. A robust method of forecasting based on fuzzy time series. *Int J Appl Math Comput* 2007;188:472–84.
- [14] Lee CHL, Lin A, Chen WS. Pattern discovery of fuzzy time series for financial prediction. *IEEE Trans Knowl Data Eng* 2006;18:613–25.
- [15] Huanrg KH, Yu THK, Hsu YW. A multivariate heuristic model for forecasting. *IEEE Trans Syst Man Cybern* 2007;37:836–46.
- [16] Yolcu U, Egrioglu E, Vedide R, Uslu R, Basaran MA, Aladag CH. A new approach for determining the length of intervals for fuzzy time series. *Appl Soft Comput* 2009;9:647–51.
- [17] Garg B, Beg MMS, Ansari AQ, Imran BM. Fuzzy time series prediction model, communications in computer and information science, vol. 141. Berlin Heidelberg: Springer –Verlag; 2011. p. 126–37.
- [18] Garg B, Beg MMS, Ansari AQ, Imran BM. Soft computing model to predict average length of stay of patient, communications in computer and information science, vol. 141. Berlin Heidelberg: Springer Verlag; 2011. p. 221–32.
- [19] Garg B, Beg MMS, Ansari AQ. Employing OWA to optimize fuzzy predictor. In: *World conference on soft computing, USA; 2011*. p. 205–11.
- [20] Garg B, Beg MMS, Ansari AQ. OWA based fuzzy time series forecasting model. In: *World conference on soft computing, Berkeley, San Francisco, CA; 2011*. p. 141–77.
- [21] Garg B, Beg MMS, Ansari AQ. Enhanced accuracy of fuzzy time series predictor using genetic algorithm. In: *Third IEEE world congress on nature and biologically inspired computing; 2011*. p. 273–8.
- [22] Garg B, Beg MMS, Ansari AQ. Employing genetic algorithm to optimize OWA-fuzzy forecasting model. In: *Third IEEE world congress on nature and biologically inspired computing; 2011*. p. 285–90.
- [23] Garg B, Beg MMS, Ansari AQ. A new computational fuzzy time series model to forecast number of outpatient visits. In: *Proc. 31st annual conference of the North American fuzzy information processing society (NAFIPS 2012); 2012*. p. 1–6.
- [24] Garg B, Garg R. Enhanced accuracy of fuzzy time series model using ordered weighted aggregation. *Appl Soft Comput, Elsevier* 2016;8:265–80.

Bindu Garg is working as associate professor in computer science department and Dean R&D of Bharati Vidyapeeth's College of Engineering, New Delhi. Her area of research and specialization are Time Series Analysis, Soft Computing, Neural networks, Fuzzy logic, Genetic Algorithm, Forecasting Applications, Analysis and Designing of Algorithm, Data Structure, Object Oriented Programming, Numerical Analysis, Cloud Computing and Big data.

Shubham Aggarwal completed his B.Tech in computer science from G.G.S.I.P University in 2017. His interests are Soft Computing, Neural Networks, Forecasting Applications and Designing of Algorithms.

Jatin Sokhal completed his B.Tech in computer science from G.G.S.I.P University in 2017. His interests are Algorithms optimization, Neural Networks, Forecasting Applications and Data Structures.